Anthony PIERRE

PhD Candidate

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Experience

Research Fellow & PhD Candidate *Paris, FR Nov 2025 – Present*
Nexialog Consulting - R&D

- Contributed to the development of a Risk-Neutral Economic Scenario Generator for market-consistent valuation and actuarial applications
- Designed NLP use cases for the insurance industry
- Preparing a dual PhD in Mathematics and Economics at Université Paris-Dauphine, focused on Temporal Graph Neural Networks for spatio-temporal clustering

R&D Mission - Flood Risk Modelling *Major European Insurer - Mission while at Nexialog* *Paris, FR Jul 2025 – Sept 2025*

- Reconstruction of continuous flood surfaces from TRI regulatory maps using quantile-based terrain anchoring and topographic data (RGE ALTI / LiDAR)
- Design of interpolation and smoothing algorithms to remove discontinuities between flood polygons and improve local hydraulic consistency
- Implementation of graph-based extrapolation methods (Dijkstra shortest-path propagation) to extend flood areas according to terrain constraints
- Conversion of hazard maps into financial losses through exposure-level impact modelling on insured assets

Quant Analyst/Developer & Actuarial Intern *Nexialog Consulting* *Paris, FR Apr 2025 – Sept 2025*

- Development of a Real World Economic Scenario Generator (ESG): implementation of stochastic models for interest rates, inflation, equities and real estate (Vasicek, Two-Factor Vasicek, Hardy regime-switching, Phillips Curve, Hull-White extensions)
- Full-stack development of an internal ESG web platform: Python, Django, HTML, CSS, JavaScript, HTMX
- Implementation of statistical validation tools: backtesting, convergence analysis, sensitivity testing, stress scenarios and scenario-consistency controls
- Drafted technical notes on Economic Scenario Generators, actuarial modeling standards and regulatory applications (Solvency II, ORSA, IFRS 17, internal models)

Consultant Data Scientist *Paris, FR Oct 2024 – Feb 2025*
Startups and SMEs

- Supported the growth of a startup by applying Machine Learning and Deep Learning techniques to climate-risk data and predictive analytics
- Developed a web application for the sales division of an agri-food SME, including an intelligent recipe search engine and data-driven commercial tools
- Leveraged 20+ years of historical client data to build customer analytics models: temporal clustering, behavioral pattern detection, forecasting and segmentation

Data Scientist Intern *Mayane Labs* *Paris, FR Apr 2024 – Sep 2024*

- Data Science under the direction of Emma Haziza (publicly renowned hydrologist), participation in the development of software using satellite data

Private Tutor in Mathematics *Complétudes* *Paris, FR Feb 2023 – Jul 2023*

Education

Ecole Polytechnique *MSC&T Second Year, Economics, Data Analytics and Corporate Finance, Finance track* *Sept 2024 – Sept 2025*

- **Coursework:** Stochastic Calculus, Financial Decisions under risk I & II, Financial Markets, Private Equity, Big Data, Corporate Finance, Econometrics of Competition, Valuation of Startups, M&A

Université Paris-Dauphine *Master Data Science for Economics, Mention Quantitative Economics, 2nd Year* *2023 – 2024*

- **Grade: High Honours, Top of the class, 17.887/20**
- **Coursework:** Machine/Deep Learning, Industrial Organization, Financial Data and Systemic Risk, Blockchain Economics, Time Series Analysis and Anomaly Detection, NLP, Algorithmic Trading

Université Paris-Dauphine *Master Quantitative Economics, 1st Year* *2022 – 2023*

- **Grade: Honours, Rank 2, 14.9/20**
- **Coursework:** Microeconomics I and II, Macroeconomics I and II, Macroeconometrics, Microeconometrics, Game Theory

Université Paris-Est Créteil *Bachelor in International Economics and Management* *2021 – 2022*

- **Grade: Honours, Valedictorian**
- **Coursework:** Econometrics, Game Theory, Microeconomics, Stata and Python, General Algebraic Systems

Projects

Investment Fund Web Platform *Collaborative Initiative*

- Centralization of fundamental, technical and market data for investment decision-making
- Implementation of portfolio optimization and allocation strategies
- Governance system including voting mechanisms, member decisions and capital allocation rules

High-Frequency Trading Bot *Personal Initiative*

- Tools Used: Advanced Statistics, Time Series Analysis, Sentiment Analysis, Neural Network (LSTM), Chart Analysis, Python
- Highly appreciated by professors

Characterization of Exoplanetary Systems *Master 2 Data Science for Economics*

- Tools Used: Advanced Statistics, Python, Semi/Non-Supervised ML/DL, R, Spatial Statistics
- **Rank: 1st**

Thesis: "Machine Learning and Econometrics to Predict Stock Market Prices" *End of Bachelor Thesis*

- Tools Used: Python, Statistics, Clustering, Econometrics
- **Rank: 1st**

Skills

Programming Languages/Softwares: Python, R, SQL, MATLAB, Stata, Solidity, QGIS, Excel, HTML, CSS, JS

Languages: French (Mother Tongue), English (Fluent), Spanish (Intermediate)